

Perception and practices towards the use of social networking sites and artificial intelligence for academic purposes among postgraduate students in various health science courses in South India

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Summary Background. Social networking sites (SNSs) provide postgraduate students with numerous opportunities for educational networking in their field of specialization. Artificial intelligence (AI) is another promising tool that has enhanced the learning experiences of postgraduate students. Despite the various advantages of incorporating these platforms and technologies in postgraduate training in education, research, and patient care, postgraduate students pursuing health science courses have expressed various concerns related to their usage.

Objectives. To assess the perceptions and practices of postgraduate students pursuing health science courses regarding the use of SNSs and AI for educational purposes, research, and patient care and to determine their determinants.

Material and methods. Data was collected via a self-administered questionnaire designed as a Google form.

Results. Of the 204 participants, 153 (75.0%) used SNSs for academic purposes. The most common benefit gained by the users concerning SNSs was faster access to information, whereas the most common drawback stated was a lack of authentic data. 158 (77.4%) participants favored its use for postgraduate training. 165 (80.9%) participants had used AI tools for academic purposes. The most common benefit gained by the users of AI was faster access to information, and the most common drawback was that it limits critical thinking. A total of 166 (81.4%) favored its usage for postgraduate training. 100 (49%) participants wanted some form of training in the use of AI applications. Place of residence and type of course were associated with the use of SNSs for academic purposes. Favorable perceptions of using SNSs and AI for academic purposes were associated with the usage of SNSs ($p < 0.001$) and AI ($p = 0.008$), respectively, among postgraduate students.

Conclusions. At least three-fourths of postgraduate students had used SNSs used SNSs or AI for academic purposes. Favorable perceptions were associated with the use of SNSs and AI. Hence, there is a need to conduct training programs among all health science postgraduate students to give them the right perspective of these platforms and to encourage their usage.

Key words: professional practice, perception, social networking, artificial intelligence, health occupations, students.

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Background

Social networking sites (SNSs) are platforms that help professionals with similar educational backgrounds and areas of interest connect [1]. Its usage as a medium of teaching-learning has become popular among medical students. A meta-analysis reported that as many as one in five medical students use SNSs for academic purposes [2]. The educational applications of SNSs are offered as structured courses at some medical colleges, although their usage is inconsistent and fragmented [3, 4]. The recently introduced competency-based medical education curriculum in India also promotes e-learning as a mode of self-directed learning [5]. This indirectly refers to the use of SNSs for medical education. The CBME curriculum also encourages

students to perform an elective on “computers and artificial intelligence in health care”.

SNS has also promoted professional development and learning among postgraduate medical students. It provides them with numerous opportunities for educational networking in their field of specialization. Access to recent updates in the fields of medical education, research, and patient care has become more accessible and faster with the help of SNSs [6].

Teachers using SNSs in their teaching activities find this opportunity a new way to connect with students. Students also find this platform a new way to retrieve and share their learning resources and assignments [7]. SNS platforms have thus become the primary means of communication among students at higher levels of education [8]. Students have also experienced



learning through these platforms easier for cross-cultural understanding and adaptation and to summarize information in familiar formats [9].

Despite the various advantages of incorporating SNSs into higher education and the extent of their usage by students and their tutors for nonformal educational purposes, there still exists a disparity between the positive perceptions of SNSs and their practical usage [10]. Serious risks in the form of violations of patient privacy and unethical behavior are associated with SNSs [11, 12]. Users struggle with issues concerning confidentiality and e-professionalism. The content shared on these platforms is unreferenced; hence, there is a question mark on the accuracy of information [12]. The potential negative impacts on mental health and being a source of distraction are other concerns reported among users [12].

Hence, appropriate guidance regarding various SNS applications and their responsible use is necessary among learners before they can be used as powerful teaching tools that promote collaborative learning that enables academic growth among both students and their tutors [9].

Artificial intelligence (AI) is another promising tool that has gained popularity. It refers to the use of computer systems to execute tasks that typically require human intelligence. Its most impactful application lies in supporting individualized learning experiences. It enhances engagement and comprehension, thereby improving the learning experiences of postgraduate students in higher educational avenues [13]. However, the use of AI is also not without disadvantages. It leads to issues related to dependence on this technology, data privacy concerns, and a lack of human touch in the services provided [14].

Exploring university websites yields more relevant and accessible information about SNS curricula and AI usage if it is unavailable in online databases. Before introducing SNSs and AI for postgraduate curriculum development, it is important to assess the prior experiences and perceptions of students towards SNSs and AI. This process helps to determine the content and teaching-learning strategies and the criteria for determining the endpoint. These aspects should be aligned with a suitable framework for enhancing the clarity and structure of the curriculum.

Prior studies on the use of SNSs for educational purposes among postgraduate medical students have presented combined findings for both undergraduate and postgraduate medical students [11, 12]. Studies performed exclusively among postgraduate students have targeted only those from non-health science backgrounds [6, 15–20]. Therefore, a study is needed to assess the perceptions and practices of health sciences PG students regarding the usage of SNSs and AI for educational purposes, research, and patient care and to find out its determinants.

Objectives

To assess the perceptions and practices of postgraduate students pursuing health science courses regarding the use of SNSs and AI for educational purposes, research, and patient care and to study their determinants.

Material and methods

This cross-sectional study was performed at a private medical college in September 2024. Institutional Ethics Committee approval was obtained on September 18, 2024. The protocol number was IEC KMC MLR 09/2024/574. Permission was obtained from the dean of this institution. All postgraduate students pursuing courses such as medicine, dentistry, physiotherapy, audiology and speech-language pathology, and clinical psychology were included in this study. Those who did not have a minimum of six months of experience as postgraduate students and those who did not consent to participate were excluded.

The sample size was calculated via the formula $n = 4pq/d^2$. p was substituted with 49%, which was the proportion of PG students who had used SNSs to share academic information in a study performed in Kashmir, India [18]. At 95% confidence intervals and 15% relative precision, the sample size was 185. Adding 10% to this number, to compensate for nonresponses, the final sample size was rounded to 205 participants. Data was collected via a self-administered questionnaire designed as a Google form. Pilot testing of the questionnaire was performed among five PGs who were not included in the main study. The information sheet and the consent form were in the initial part of the questionnaire. The participants who were not willing to participate were given the option to decline consent, which led to the termination of participation in this study. The first section of the questionnaire inquired about the sociodemographic details of the participants. The second section asked about their perceptions and practices of using SNSs for academic purposes, research, and patient care. The third section asked about the same information in relation to AI.

The questionnaire was validated with the help of two expert subjects from the Department of Medical Education of this institution.

This research study replicates the process documentation on module development for SNS and AI implementation in postgraduate training on the basis of the PARE (preparation, action, reflection, and evaluation) model [21]. Currently, the initial two steps of preparation and action are emphasized. The data and insights from the reflection and evaluation stages will be incorporated following the implementation of the module among postgraduate students in further research studies.

The data was entered and analyzed via the Statistical Package for the Social Sciences version 25. The Cronbach's alpha value of the reliability of the questionnaire was found to be 0.868. Descriptive statistics are expressed as the means, standard deviations, and percentages. Statistical tests such as the Chi-square test were used to test associations in univariate analysis. All the variables associated with the usage and perception of SNSs and AI for academic purposes among postgraduate students at a p -value of 0.2 or less were introduced into the binary logistic regression analysis. $P < 0.05$ was considered a statistically significant association.

Results

A total of 205 postgraduates completed the Google form. Among them, 204 gave consent for participation.

The mean age of the participants was 26 (SD 2.9) years. The majority of the participants were females [132 (64.7%)] (Table 1).

Table 1. Sociodemographic distribution of the study participants ($n = 204$)

Characteristics	Number	Percentage
Age group (years)		
20–22	33	16.2
23–25	43	21.1
26–28	96	47.1
29–31	28	13.7
≥ 32	4	1.9
Gender		
Females	132	64.7
Males	72	35.3
Specialty		
Clinical subjects*	78	38.2
Para-clinical subjects**	20	9.8
Preclinical subjects ***	6	3.0
Dentistry****	32	15.7

Table 1. Sociodemographic distribution of the study participants (n = 204)

Characteristics	Number	Percentage
Allied Health Sciences*****	68	33.3
Year of study		
First year	98	48.0
Second year	66	32.4
Third year	40	19.6
Marital status		
Unmarried	178	87.2
Married	24	11.8
Widow	1	0.5
Separated	1	0.5
Permanent place of residence		
Urban	153	75.0
Semiurban	44	21.6
Rural	7	3.4
Total	204	100.0

*General Surgery 16, Obstetrics and Gynecology 10, General Medicine 9, Pediatrics 9, Orthopedics 7, Ophthalmology 6, Radiodiagnosis 6, Dermatology 5, Otorhinolaryngology 4, Emergency Medicine 3, Psychiatry 2, Anesthesia 1; **Community Medicine 8, Pathology 7, Forensic Medicine 2, Pharmacology 2, Microbiology 1; ***Anatomy 2, Biochemistry 2, Physiology 2; ****Conservative Dentistry and Endodontics 9, Periodontics 7, Oral Medicine and Radiology 5, Preventive Dentistry 3, Oral Surgery 2, Orthodontics and Dentofacial Orthopedics 2, Prosthodontics and Crown and Bridge 2, Oral Pathology and Microbiology 1, Pediatric Dentistry 1; *****Audiology and Speech-Language Pathology 27, Clinical Psychology 19, Physiotherapy in Cardiopulmonary Sciences 8, Physiotherapy in Neurosciences 5, Physiotherapy in Musculoskeletal Sciences 4, Physiotherapy in Pediatrics 2, Physiotherapy in Sports Medicine 2, Physiotherapy in Critical Care Rehabilitation 1.

As many as 153 (75.0%) participants used SNSs for academic purposes. Among them, most users used the same platform for medical education purposes [130 (85.0%)], and YouTube was the most commonly used platform. WhatsApp was the medium most commonly used by postgraduate students for research, patient care, and the sharing of educational materials (Table 2).

Table 2. Usage characteristics of social networking sites for academic purposes among postgraduate students in various health science courses (n = 204)

Characteristics	Number	Percentage
Had used social networking sites for academic purposes		
Yes	153	75.0
No	51	25.0
Reasons for non-usage (n = 51)*		
Unfamiliarity	41	80.4
Disinterest	19	37.2
Source for distraction	13	25.5
Not a reliable medium	2	3.9
Type of academic purposes for which it was used (n = 153)*		
Medical education	130	85.0
Research	107	69.9
To discuss patient care	87	56.9
Type of SNS used for educational purposes (n = 130)*		
YouTube	130	100.0
WhatsApp	90	69.2

Table 2. Usage characteristics of social networking sites for academic purposes among postgraduate students in various health science courses (n = 204)

Characteristics	Number	Percentage
Instagram	90	69.2
Telegram	81	62.3
Google Meet	69	53.1
Microsoft Teams	66	50.8
Marrow	61	46.9
LinkedIn	56	43.1
Zoom	56	43.1
UpToDate	51	39.2
Unacademy	39	30.0
DailyRounds	33	25.4
PrepLadder	30	23.1
Facebook	29	22.3
Pinterest	26	20.0
Reddit	21	16.1
Twitter	21	16.1
Blackboard	15	11.5
Snapchat	14	10.8
Sermo	11	8.5
Spotify	8	6.1

SNS used to share educational material like class notes and scientific literature (n = 130)*

WhatsApp	119	91.5
Telegram	82	63.1
Microsoft Teams	74	56.9
YouTube	70	53.8
Marrow	49	37.7
PrepLadder	39	30.0
Others#	2	1.5

Type of SNS used for research (n = 107)*

WhatsApp	82	76.6
LinkedIn	61	57.0
Facebook	26	24.3
Telegram	5	4.7
Instagram	4	3.7
Others##	5	4.7

Type of SNS used for communicating with peer groups/senior doctors/faculty members regarding patient care (n = 87)*

WhatsApp	87	100.0
LinkedIn	46	52.9
Facebook	17	19.5
Telegram	4	4.6
Instagram	4	4.6
Microsoft Teams	2	2.3
Twitter	1	1.1

Frequency of usage of various SNS for academic purposes (n = 153)

More than once a day	61	39.9
Once a day	42	27.4
Once a week	43	28.1
Once a month	7	4.6

*Multiple responses; #Google meet 1, Instagram 1; ##UpToDate 2, YouTube 2, Twitter 1.

Table 3. Perception of the usage of social networking sites for academic purposes among postgraduate students in various health science courses (n = 204)

Characteristics	Number	Percentage
Benefits gained by using SNS for various academic purposes (n = 153)*		
Faster access to information	131	85.6
Improvement in clinical knowledge	118	77.1
Educational networking with others	112	73.2
Enhanced attention span	49	32.0
Others#	4	2.6
Drawbacks experienced while using this platform for academic purposes (n = 153)*		
Lack of authentic data	100	65.4
Lack of seriousness in group discussions	75	49.0
Privacy issues	71	46.4
Wastage of time	53	34.6
Others##	8	5.2
Challenges experienced while using this platform for academic purposes (n = 153)*		
Eye strain	100	65.4
Slow internet	68	44.4
Addiction	66	43.1
Headache	63	41.2
Accessibility issues	62	40.5
Ethical issues	50	32.7
Lethargy	49	32.0
Incorrect information	2	1.3
Usage of SNS for academic purposes should be recommended for postgraduate training		
Strongly agree	59	28.9
Agree	99	48.5
Uncertain	37	18.2
Disagree	8	3.9
Strongly Disagree	1	0.5
Reasons for strong agreement or agreement (n = 158)*		
Allows easier interaction with facilitators	133	84.2
Facilitates collaborative learning	111	70.2
Provides up-to-date information	103	65.2
Others@	3	1.9
Need of training for using SNS for academic purposes (n = 204)		
Specify for which SNS training is required (n = 77)*	77	37.7
Facebook	32	41.6
WhatsApp	27	35.1
Twitter	20	26.0
LinkedIn	20	26.0
PrepLadder	9	11.7
Marrow	7	9.1

*Multiple responses; #Better understanding of concept 2, greater access to information 1, improved time management 1; ##Health problems associated with overuse of SNS 2, Sleep disturbances 2, Source of stress 1, The multitude of sites leads to distraction, thereby making it difficult to gather information 1, Readers stop referring to standard textbooks owing to readily available information on these platforms 1, Screenshot increases at the expense of interaction time with people around 1; @Popularity of these platforms 1, Easier access of information 1, Quick access of information 1.

The most common benefit gained by using SNSs for academic purposes was faster access to information, as stated by 131 (85.6%) of the 153 postgraduate students. The most common drawback was the lack of authentic data, as stated by 100 (65.4%) of the 153 postgraduate students. As many as 158 (77.4%) of the total participants strongly agreed or agreed that SNSs for academic purposes should be recommended for postgraduate training. However, 9 participants disagreed or strongly disagreed with the others (Table 3). The reasons for this are that it provides a lot of erroneous information (stated by 8), there

is a lack of seriousness in the discussions on these platforms (stated by 6), several spam posts (stated by 4), ethical concerns (stated by 3), misguides youth (stated by 2), and more of an advertising platform relying on more views and likes for functioning (stated by 1).

As many as 165 (80.9%) postgraduate students had used AI tools for academic purposes. Among them, most users used these AI tools for research purposes [133 (80.6%)], and ChatGPT was the most commonly used application [158 (95.8%)] (Table 4).

Table 4. Usage characteristics of artificial intelligence applications for academic purposes among postgraduate students in various health science courses (n = 204)

Characteristics	Number	Percentage
Had used AI applications for academic purposes		
Yes	165	80.9
Type of academic purposes for which AI was used (n = 165)*		
Research	133	80.6
Medical education	130	78.8
To discuss patient care	65	39.4
Specify the AI applications used (n = 165)*		
ChatGPT	158	95.8
Meta AI	67	40.6
Gemini	43	26.1
Claude	26	15.8
Machine Learning Algorithms	24	14.5
CoPilot	2	1.2
QuillBot	2	1.2
Activity for which AI was used (n = 165)*		
For research manuscript preparation	50	30.3
To get recent updates in patient care	43	26.1
For content creation	40	24.2
For information on treatment algorithms	33	20.0
For the review of literature search	20	12.1
Paraphrasing	16	9.7
Content interpretation	13	7.9
For the graphical presentation of the results	12	7.3
Plagiarism check	8	4.8
For taking down notes	3	1.8
Others#	6	3.6
Frequency of usage of various AI applications for academic purposes (n = 165)		
More than once a day	44	26.7
Once a day	24	14.5
Once a month	44	26.7
Once a week	53	32.1

*Multiple responses; #For writing discharge summaries 1, framing questions 1, manuscript editing 1, for information on rehabilitation protocols 1, reference writing 1, finding good journals 1.

The most common benefit gained by using AI for academic purposes was faster access to information, as stated by 123 (74.5%), and the most common drawback stated was that it limits critical thinking, as stated by 97 (58.8%) out of 165 postgraduate students. As many as 166 (81.4%) of the total participants strongly agreed or agreed that AI applications should be recommended for academic purposes involving postgraduate

training (Table 5). However, disagreements or strong disagreements were expressed by 6 participants. The reasons stated were that AI leads to overreliance on technology (by 5), has the potential for bias (by 4), reduces critical thinking (by 3), has privacy concerns (by 2), and lacks human interaction (by 2). 100 (49%) participants wanted some form of training in the use of AI applications (Table 5).

Table 5. Perception of the use of artificial intelligence applications for academic purposes among postgraduate students in various health science courses (n = 204)

Characteristics	Number	Percentage
Benefits gained by using AI applications for various academic purposes (n = 165)*		
Faster access to information	123	74.5
Enhanced research efficiency	108	65.4
Simplification of difficult-to-understand concepts	103	62.4
Easier analysis of research data	87	52.7
Drawbacks experienced while using this platform for academic purposes (n = 165)*		
Limits critical thinking	97	58.8
Misinterpretation risks	94	57.0
Overreliance on tools	86	52.1
Quality inconsistency	77	46.7

Table 5. Perception of the use of artificial intelligence applications for academic purposes among postgraduate students in various health science courses (*n* = 204)

Characteristics	Number	Percentage
Privacy concerns	59	35.8
Challenges experienced while using this platform for academic purposes (<i>n</i> = 165)*		
Understanding tool limitations	75	45.4
Handling inaccurate outputs	75	45.4
Ensuring data privacy	74	44.8
Integrating with existing systems	62	37.6
Navigating tool complexity	48	29.1
The usage of AI applications for academic purposes should be recommended for postgraduate training (<i>n</i> = 204)		
Strongly agree	71	34.8
Agree	95	46.6
Uncertain	32	15.6
Disagree	3	1.5
Strongly disagree	3	1.5
Reasons for strong agreement or agreement (<i>n</i> = 166)*		
AI enhances research efficiency	131	78.9
It gives personalized learning support	109	65.7
It has streamlined administrative tasks	88	53.0
It has data-driven insights	85	51.2
It improves decision-making skills	72	43.4
Improves understanding	3	1.8
Saves time	1	0.6
Need of training for using AI for academic purposes (<i>n</i> = 204)	100	49.0
Specify AI for which AI training is required (<i>n</i> = 100)		
ChatGPT	8	8.0
Gemini	2	2.0
Any AI tool	90	90.0
Specify the area of training required (<i>n</i> = 100)*		
Dissertation writing	39	39.0
Review of literature search	34	34.0
Preparation of manuscript before sending it for publication	30	30.0
Searching scientific information	19	19.0
Paraphrasing the text	18	18.0
Ethical issues pertaining to the usage of AI	15	15.0
Making PowerPoint slides for the conference presentation of the article	12	12.0

Some of the open comments given by the participants were as follows: AI needs to be integrated into postgraduate medical education to accelerate research and support medical education. This will save considerable time and effort for postgraduate students, which can be utilized for patient care (stated by 1). AI tools are currently needed for every postgraduate student to help them write and publish research articles. It will, therefore, encourage them to do more research work (stated by 1). AI and SNS usage are rapidly flourishing these days, and this should help in patient care (stated by 1). It is okay to slow the pace and learn critically and naturally instead of relying on AI, which is tied to bias and various other ethical issues. Often, ChatGPT learning may induce overdependence and limit critical thinking, which may leave us unprepared for real-world challenges (stated by 1). The accuracy of the available information in various AI platforms needs to be cross-checked before it is used in research communications (stated by 1). Compared with hospital-based and classroom teaching, AI tools are not effective. These tools make the work easy on paper. Personal doctor–patient interactions cannot be replaced by these tools, as they lack a human touch,

which is the essential core of this noble profession (stated by 1). Overdependence on AI tools in medical education can lead to reduced critical thinking (stated by 1).

In the binary logistic regression analysis, usage of SNSs for academic purposes was significantly greater among postgraduate students whose permanent residence was in urban areas ($p = 0.03$), those pursuing AHS courses ($p < 0.001$), those with MD/MS clinical specialties ($p = 0.028$), and those who strongly agreed/agreed in favor of using SNSs for academic purposes ($p < 0.001$) (Table 6).

According to the binary logistic regression analysis, the use of AI for academic purposes was significantly greater among postgraduate students who strongly agreed/agreed in favor of using it for academic purposes for postgraduate training ($p = 0.008$) (Table 6).

According to the univariate analysis, the marital status of the participants was associated with their perceptions of the use of SNSs ($p = 0.017$) and AI applications ($p = 0.049$) for academic purposes during postgraduate training (Table 7).

Table 6. Association between variables and usage of SNSs and AI for academic purposes among postgraduate students in various health science courses (n = 204)

Characteristics	Used SNS for academic purposes No. (%)	Did not use SNS for academic purposes No. (%)	Total	UOR (95% CIs)	Chi-square value, p-value	AOR (95% CIs)	p
Age group (years)							
20–25	61 (80.3)	15 (19.7)	76	1.591 (0.803, 3.153)	1.789, 0.181	1.109 (0.445, 2.763)	0.825
> 25	92 (71.9)	36 (28.1)	128	1		1	
Permanent place of residence							
Urban	119 (77.8)	34 (22.2)	153	1.75 (0.873, 3.509)	2.519, 0.113	2.377 (1.087, 5.197)	0.03
Semiurban/Rural	34 (66.7)	17 (33.3)	51	1		1	
Course							
Allied Health Science Specialties	58 (85.3)	10 (14.7)	68	5.118 (1.949, 13.441)	12.078, 0.007	9.67 (3.204, 29.188)	< 0.001
MD/MS in clinical specialties	59 (75.6)	19 (24.4)	78	2.137 (0.714, 6.394)		2.908 (1.125, 7.518)	0.028
MD/MS in pre/para-clinical specialties	19 (73.1)	7 (26.9)	26	1.868 (0.801, 4.358)		2.112 (0.616, 7.240)	0.234
MDS courses	17 (53.1)	15 (46.9)	32	1		1	
Perception regarding the use of SNS for academic purposes							
Strongly Agree/Agree	130 (82.3)	28 (17.7)	158	4.643 (2.288, 9.421)	19.797, < 0.001	7.065 (3.144, 15.873)	< 0.001
Uncertain/Disagree/Strongly disagree	23 (50.0)	23 (50.0)	46	1		1	
Total	153	51	204				
Characteristics	Used AI for academic purposes No. (%)	Did not use AI for academic purposes No. (%)	Total	UOR (95% CIs)	Chi-square value, p-value	AOR (95% CIs)	p
Year of study							
Second year	58 (87.9)	8 (12.1)	66	2.48 (1.04, 5.91)	5.118, 0.077	2.3 (0.763, 6.936)	0.139
Third year	34 (85.0)	6 (15.0)	40	1.941 (0.729, 5.168)		0.884 (0.271, 2.877)	0.837
First year	73 (74.5)	25 (25.5)	98	1		1	
Course							
MDS courses	27 (84.4)	5 (15.6)	32	3.375 (0.978, 11.65)	7.228, 0.065	3.465 (0.944, 12.72)	0.061
Allied Health Science Specialties	57 (83.8)	11 (16.2)	68	1.08 (0.351, 3.326)		1.161 (0.345, 3.908)	0.809
MD/MS in clinical specialties	65 (83.3)	13 (16.7)	78	1.042 (0.329, 3.298)		0.799 (0.244, 2.618)	0.711
MD/MS in pre/para-clinical specialties	16 (61.5)	10 (38.5)	26	1		1	
Perception regarding the use of AI for academic purposes							
Strongly Agree/Agree	141 (84.9)	25 (15.1)	166	3.29 (1.501, 7.209)	9.488, 0.002	3.127 (1.344, 7.275)	0.008
Uncertain/Disagree/Strongly disagree	24 (63.2)	14 (36.8)	38	1		1	
Total	165	39	204				

Table 7. Association between variables and perceptions of SNSs and AI for academic purposes among postgraduate students in various health science courses (n = 204)

Characteristics	Perception regarding the use of SNS for academic purposes Strongly agree/ Agree No. (%)	Perception regarding the use of SNS for academic purposes Uncertain/Disagree/ Strongly Disagree No. (%)	Total	UOR (95% CIs)	Chi-square value, p-value	AOR (95% CIs)	p
Marital status							
Unmarried/Widowed/ Separated	144 (80.0)	36 (20.0)	180	2.857 (1.173, 6.957)	5.692, 0.017		
Married	14 (58.3)	10 (41.7)	24	1			
Total	158	46	204				
Characteristics	Perception regarding the use of AI for academic purposes Strongly Agree/ Agree No. (%)	Perception regarding the use of AI for academic purposes Uncertain/Disagree/ Strongly Disagree No. (%)	Total	UOR (95% CIs)	Chi-square value, p value	AOR (95% CIs)	p
Gender							
Male	63 (87.5)	9 (12.5)	72	1.97 (0.88, 4.43)	2.756, 0.097	1.278 (0.601, 2.718)	0.525
Female	103 (78.0)	29 (22.0)	132	1		1	
Marital status							
Unmarried/Widowed/ Separated	150 (83.3)	30 (16.7)	180	2.5 (0.98, 6.37)	3.881, 0.049	0.92 (0.319, 2.656)	0.877
Married	16 (66.7)	8 (33.3)	24	1		1	
Total	166	38	204				

Discussion

In the present study, three-fourths of the postgraduate students had used SNSs for academic purposes. In other studies, done in Bangladesh [16] and South Africa [15] among postgraduate students in non-health sciences courses, 89.3% and 94% of them, respectively, had used SNSs for academic purposes. When used for educational purposes, SNSs improve learners' critical thinking and problem-based learning skills; hence, their usage needs to be promoted among students pursuing higher education [22–24].

The most common SNSs used by the postgraduate students in the present study for educational purposes were YouTube, and for research purposes, and to communicate with other doctors regarding patient care was WhatsApp. In previous studies done among postgraduate students in non-health sciences courses, WhatsApp was the most common platform used for academic purposes [15, 20]. The high usage of WhatsApp by students may be due to low data consumption, making it the most available, accessible, and affordable platform for educational purposes. It also makes the data sharing of large files straightforward and fast. YouTube, which is the world's largest storehouse of educational videos, is popular among students. A three-dimensional depiction makes understanding difficult concepts easier for students [25]. Moreover, searching for a specific topic on YouTube is easier without the need to have an account to log into for usage, as is required in most other SNSs.

The proportions of various postgraduate students using SNS platforms for academic purposes were as follows: Zoom [by 79% [15]], WhatsApp [by 31.4% [20] and 83% [15]], Facebook [by 14% [17]], YouTube [by 20% [17], 24.3% [20], and 67% [15]], blogs [by 15% [17]], Wikis [by 30% [17]], Podcasts [by 3% [17]], Twitter [by 12% [17]], LinkedIn [by 6% [17]], Telegram [by 10% [15]], and Quora [by 6% [15]]. The use of various types of SNSs for academic purposes by students around the world in higher education highlights their role in enhancing the learning engagement

of students in their assignments and research projects outside of college hours. Students also tend to connect to several SNSs for more educational networking opportunities and to develop higher levels of cognitive skills in their disciplines [15].

Additionally, the COVID-19 outbreak has led to the popularity of platforms such as Zoom for organizing online classes and conferences. Services such as blogging, audio chat, and video chat are other features of SNSs, making them useful tools for educational purposes.

The majority of the postgraduate students who used SNSs for academic purposes in this study used them mainly for medical education. Previous studies reported that SNSs were used by postgraduate students to share information [40% [20], 49% [18], and 86% [19]], to attend seminars and conferences [24.7% [18]], to read and access information [8.8% [18], 22.9% [20], and 62% [17]], to complete assignments [81.2% [15]], to prepare for exams [50% [15]], to communicate with facilitators [28% [17] and 31.4% [20]], to engage in group discussions with peer groups [66.7% [15], and to support professional development and lifelong learning experiences [43% [17] and 95.8% [15]]. SNSs, in addition to their use for educational purposes, were used by postgraduate students in the present study for research purposes and for patient care. In other studies, 1.4% [20] and 87.5% [15] of postgraduate students with a non-health sciences background used SNSs for research purposes.

The most common reason stated by postgraduate students who did not use SNSs for academic purposes in the present study was unfamiliarity with SNS platforms. In a study conducted in South Africa, the reasons stated by three postgraduate students for not using social media for academic purposes were a lack of authenticity of information (stated by two) and a lack of institutional support in training on its usage (stated by one) [15].

In the present study, the majority of postgraduate students perceived faster access to information, improvement in clinical knowledge, and educational networking with others as benefits of using SNSs for various academic purposes. In prior studies, postgraduate students reported that the use of SNSs

for academic purposes improved their academic performance [stated by 74.3% [6]]; helped in instant communication [stated by 70.2% [6] and 92.7% [16]]; made them feel easy in expressing themselves [stated by 50.3% [6]]; strengthened individual well-being and self-esteem [stated by 45.6% [6]]; enhanced communication between learners and instructors [stated by 91.4% [20] and 98.7% [16]]; built relationships between learners and instructors [stated by 100% [16]]; enhanced research networking [stated by 60.8% [6]]; enhanced educational networking [stated by 56.7% [6]]; was easy to use for educational purposes [stated by 52% [17]]; added to information given through conventional learning methods [stated by 88.6% [20]]; was beneficial for advanced learning in the specialty [stated by 94.3% [20]]; helped them gain up-to-date information [stated by 98% [16]]; enhanced the thinking capacity of learners [stated by 82.9% [20]]; improved the effectiveness of group discussions [stated by 98.7% [16]]; provided easy access to job-related information [stated by 98.7% [16]]; and promoted independent learning [stated by 95.7% [20]]. From the observations of the present study and prior studies, it can be inferred that learners have a positive attitude toward the benefits of SNSs for academic purposes. It was also perceived as a medium to establish a strong relationship with other learners and instructors available on these platforms.

With respect to drawbacks related to the usage of SNSs for academic purposes, most participants in the present study perceived the lack of authentic data available on these platforms as the major deterrent. Hence, clear usage guidelines and policies to ensure professionalism and ethical considerations in relation to its usage are essential before it can be applied for academic purposes. The material posted on these platforms needs to be evaluated for accuracy and to be suitably referenced.

Health issues such as eye strain, addiction, headache, and lethargy, along with slow internet, were potential challenges reported among the several participants who had used SNSs for academic purposes in this study. In studies performed among postgraduate students from non-health science backgrounds, unreliable internet [stated by 67.8% [6], and 77.5% [18]], lack of training [stated by 48.0% [6]], lack of management support [stated by 34.5% [6]], and privacy issues [stated by 10.7% [18]] were the challenges listed by the participants in applying SNSs for academic purposes.

More than one-third of the participants in this study felt that training programs in the use of SNSs for academic purposes are required at the institutional level. In a study performed in Tanzania among postgraduate students with a non-health sciences background, 67.8% of the participants felt the same [6]. This indicates the need for training programs to guide postgraduate students in the proper use of SNSs for academic purposes.

More than 80% of the postgraduate students in this study used AI for academic purposes, and most used it for research purposes. The benefits such as faster access to a review of literature on the topic of research and its features such as drafting a full manuscript, thereby improving research efficiency in dissertation writing, were stated by most participants in this study. However, more than half of the users were concerned that it

could lead to overreliance on these applications, therefore limiting the critical thinking of the human mind. Additionally, the research output that is generated on these applications should be proofread again by the users, as the tool itself may misinterpret observations [26].

Nearly half of the participants wanted some form of training in the usage of AI applications, and the most common areas that required training, as stated by them, were related to dissertation report writing. Hence, there is a need to conduct workshops on how to use AI for research project report writing among postgraduate students in this setting.

Postgraduate students who were permanent residents from urban areas were found to use SNSs more for academic purposes in this study. This might be because of the greater popularity of SNSs and better resources available to support their usage in urban parts of India. Postgraduate students pursuing AHS courses and clinical specialties in medicine were observed to use SNSs more for academic purposes than students in other courses, probably because the clinical updates were readily available and easily transmissible on these platforms. Favorable perceptions of using SNSs and AI for academic purposes were also associated with their respective usage among postgraduate students in this setting. Hence, there is a need to create a positive perspective of SNSs and various AI applications among postgraduate students regarding their benefits for academics purposes.

Limitations of the study

The findings of this study may not be generalizable to health science postgraduate students from other settings. There may be a possibility that the participants did not report information regarding their SNSs and AI usage for academic purposes. The modules on SNSs and AI have not yet been formally implemented in postgraduate training. Hence, the evaluation data regarding students' or faculty members' reactions to this module could not be obtained at this stage of this study.

Conclusions

At least three-fourths of the postgraduate students had used SNSs or AI for academic purposes during their training period. Additionally, more than three-fourths of them favored its usage. Favorable perceptions were associated with their use of SNSs and AI. Hence, there is a need to conduct training programs among all health science postgraduate students to give them the right perspective on these platforms and to encourage appropriate usage of SNSs and AI during the training period. The opportunities for integrating these platforms in higher education must be explored to take postgraduate education in the health sector to the next level. The anticipated drawbacks and challenges, as stated by the participants, need to be addressed during training programs before the integration process.

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